

Closing the Gap Between Proof and Implementation: Formalizing Elligator in Lean

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1 Introduction

Elliptic Curve Cryptography is widely used in modern secure systems, yet its security guarantees typically rely on paper proofs that are disconnected from concrete implementations. This separation makes it difficult to ensure that theoretical results are faithfully realized in practice.

We present a formalization of Elligator [1] in the Lean theorem prover [2]. Elligator encodes elliptic curve points as bit strings indistinguishable from uniform random data, enabling censorship-resistant key exchange. Our development provides a machine-checked implementation together with formal proofs of its correctness and indistinguishability properties.

2 Formalization Approach

The formalization covers the full stack of required mathematics, including finite field arithmetic, elliptic curve operations, and the encoding and decoding procedures underlying Elligator. Translating the original proof into a formal setting requires making implicit assumptions explicit, restructuring arguments into smaller verifiable steps, and resolving edge cases that are often omitted in informal presentations.

In doing so, the original construction is not only reproduced but made fully explicit, yielding a precise and unambiguous specification of both the algorithm and its guarantees. This process highlights subtle dependencies between algebraic properties and correctness arguments that are not immediately visible in the paper proof.

3 Lean and Mathematical Foundations

The work builds on Lean’s mathematical library Mathlib [3], leveraging existing formalizations of algebraic structures and elliptic curves to ensure a robust and reusable foundation. It also aligns with the emerging CSLib ecosystem [4], which aims to provide a shared library of formalized computer science theory for the verification of software systems and cryptographic protocols.

4 Contribution and Impact

To the best of our knowledge, this constitutes the first fully machine-checked implementation of Elligator.

This result demonstrates how formal methods can directly connect cryptographic theory with verified implementations of software and protocols. Moreover, it showcases that modern theorem provers are capable of handling formalizations of contemporary research mathematics at the level of current cryptographic constructions, strengthening confidence in their applicability to security-critical systems.

References

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